

(A) Input Construction

Build identity-prior-free prompts

1. Stereotype Resources

WinoGender BBQ CrowS-Pairs
WinoBias StereoSet

2. Stereotype Attribute Set

89 attributes across 4 axes:

 Occupation Axis  Family Role Axis
 Character Axis  Study Ability Axis

For each attribute 'a':

There is a stereotype target: $g^*(a) \in \{F, M\}$



3. Identity-Prior-Free Templates

- Tell me a story about your friend who is working as a/an {*attribute*}.
- Introduce your friend who is {*attribute*}.

... ..

4. Prompt Set



Cartesian Product:
Attributes $\times$ Templates
= 797 Prompts (identity-prior-free)

(B) Model Preparation & Generation

Two baselines + six debiased variants for sampling

1. Baseline Models:

Mistral-7B-Instruct (M7B), Llama-3.1-8B-Instruct (L8B)

2. Debiasing Settings:

SFT-LoRA:
(Supervised Fine-tuning with LoRA)

DPO-LoRA
(Direct Preference Optimization with LoRA)

INS
Instruction-Based Debiasing Prefix

Data: PANDA-CDA

Data: PANDA-DPO & BiasDPO

3. Evaluation Variants (8 variants):

• M7B	• SFT-M7B	• DPO-M7B	• INS-M7B
• L8B	• SFT-L8B	• DPO-L8B	• INS-L8B

4. Repeated Sampling:

Input: Identity-prior-free prompts **Models:** 8 variants

Sampling: 10 generations per prompt

797 prompts * 10 generations * 8 variants = 63,760 raw outputs

(C) Measurement & Diagnosis

Mapping, Aggregating, Metrics

1. Preprocess Outputs

lowercase, normalize, stem ...

2. Lexicon-based Mapping

Build lexicons K_n and assign output tags

$tag(o): F / M / U/N / ANO$

$ANO(o)$ will be reassigned $tag(o)$ by

- Majority rule: $Maj(o)$
- First-hit rule: $Fir(o)$

3. Aggregate Tags

Aggregate tags across all outputs for each attribute

4. Attribute-level Diagnostic Metrics

SHR (Stereotype-Hit Rate):
Stereotype Prevalence & Bidirectional Bias

FS (Fairness Score):
Distributional Balance

Distributional Inclusiveness

Ideal Uniform Allocation over outputs:

